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DIGITAL EGYPT PIONEERS INITIATIVE — DEPI

DEPI Industry Course (Round Five)

Applied AI Agents for Education and Teaching Professional

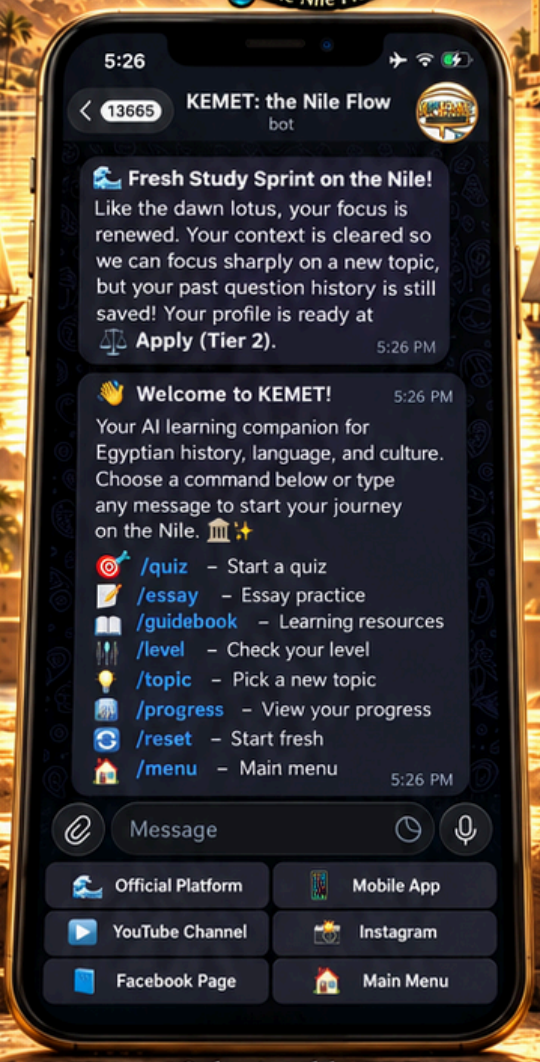
CAPSTONE PROJECT

KEMET: The Nile Flow

Telegram Chatbot UI / UX Portfolio

Adaptive Learning Conversation Experience Design

KEMET
AI Language Assistant
The Nile Flow



Submitted by

Team 136 • Group: GIZ5_IND1_G1

Supervised by: Engineer Alaa Samir

October 2026

PART I / CHATBOT UX

The conversational design brief

Why Telegram is the front door, and what the experience really contains.

DESIGN CHALLENGE

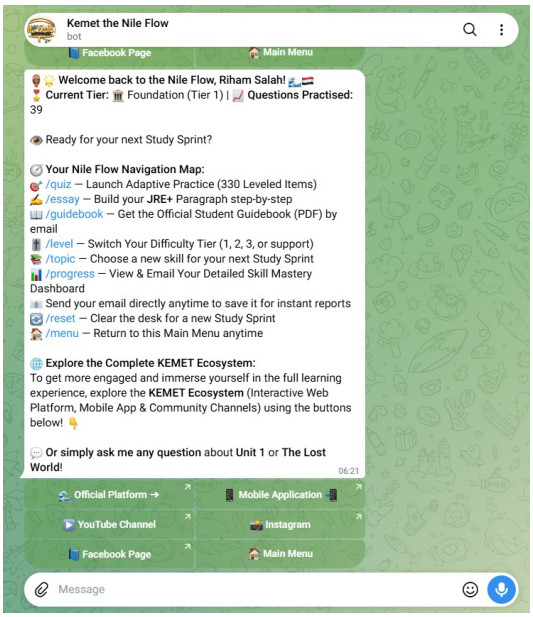
Make Grade 11 English practice accessible in a familiar chat, without burying students in menus, and while keeping KEMET's official identity intact.

DESIGN RESPONSE

Natural-language help, explicit slash commands, low-friction inline options, guided correction and one-tap routes to the wider ecosystem.

DELIVERY CONTEXT

Telegram bot linked to an n8n flow, Gemini tutoring, Qdrant data and Gmail. The Telegram chat is not a separate custom mobile application.



Evidence: Real Telegram screenshots establish the visual interface. Additional states in this document are labelled **code-derived design reconstructions**, not screenshots of tested production sessions.

PART I / CHATBOT UX

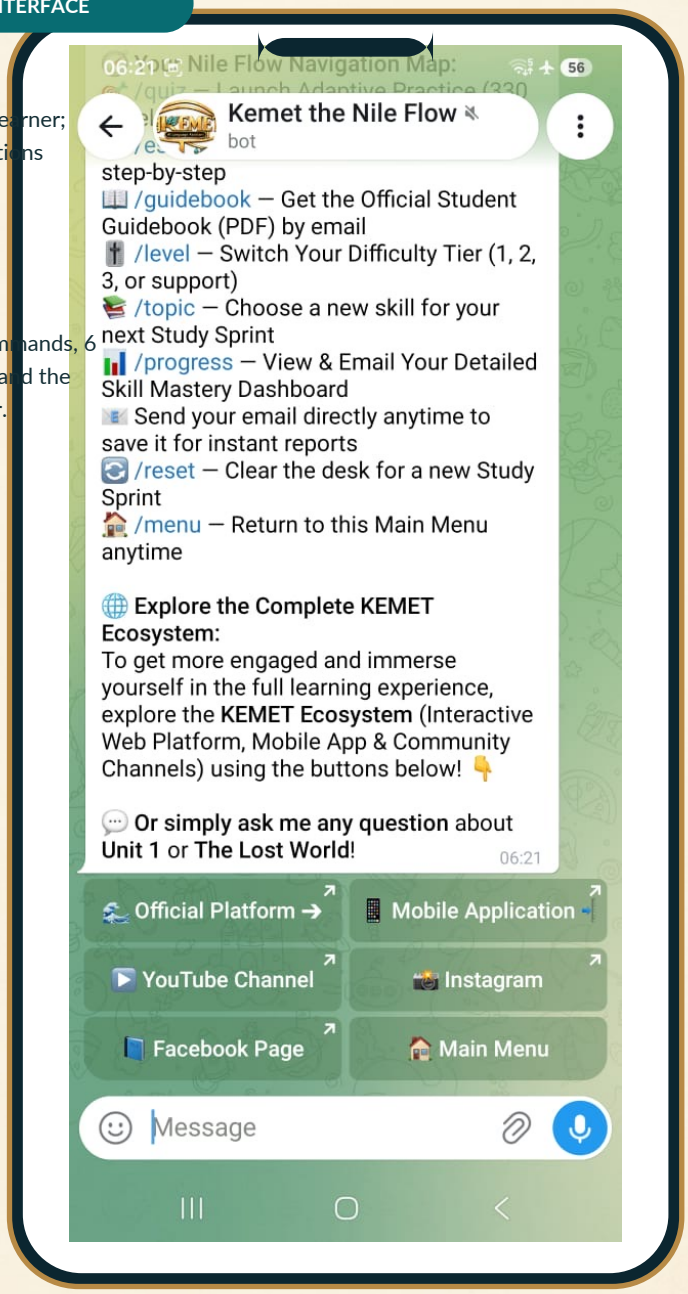
Real interface - unchanged

The supplied live Telegram screenshot is presented without replacing its menu, header, buttons or banner.

OBSERVED USER INTERFACE

Observed values: returning learner; Foundation (Tier 1); 39 questions practised.

Observed controls: slash commands, 6 external/navigation buttons and the Telegram message composer.



TELEGRAM

The green message area, header, input row and buttons belong to Telegram, not a custom KEMET mobile UI.

Design principle: preserve Telegram familiarity; present adaptive intelligence through conversation instead of recreating Telegram as a fictional app.

PART I / CHATBOT UX

Navigation: exact commands

The blue command links are the primary interaction language; not all commands are bottom-row buttons.

/quiz	Adaptive MCQ practice; question bank scroll and reply callbacks.
/essay	Start guided JRE+ paragraph creation through the tutor agent.
/guidebook	Request the official Guidebook PDF through registered email.
/level	Choose Support (0), Foundation (1), Apply (2) or Challenge (3).
/topic	Switch skill bank (V1, V2, R1, R2, P1, C1, C2, S1, L1, N1).
/progress	Show skill breakdown and prepare an email report when email exists.
/reset	Reset current study sprint while retaining cumulative history.
/menu	Pause and return to the navigational menu.

Additional recognized commands: /start, /help, /stop, /exit, /email and /guide. Plain email addresses can be accepted directly. The screenshot shows the principal eight study commands, not these helper aliases.

PART I / CHATBOT UX

Six real inline menu buttons

These actions are visible beneath the Telegram menu. The official KEMET links are not invented application screens.

Official Platform

https://glowing-tabletop-production.weweb.io/

Mobile Application

https://kemet-master-ancient-egyptian-scribe-gold-design.ai.studio/

YouTube Channel

https://www.youtube.com/@KemetTheNileFlow

Instagram

https://www.instagram.com/kemethenile/

Facebook Page

Official KEMET Facebook page, configured in JSON

Main Menu

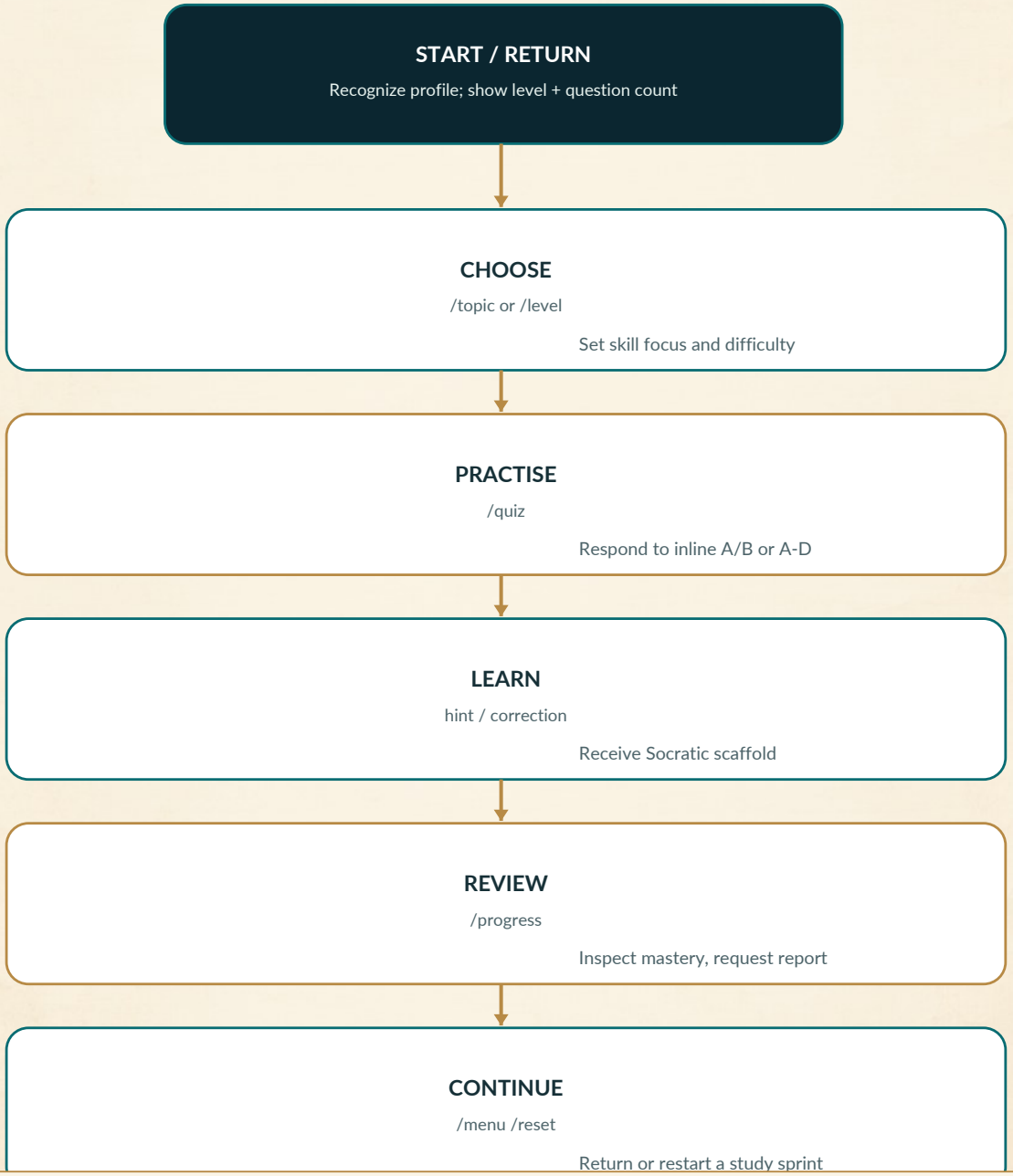
Telegram callback -> intended menu route

Important functional detail: Main Menu is a **callback**, whereas the other five are URL links. One callback value in the supplied JSON contains extra text instead of plain **menu**; this must be tested and corrected.

PART I / CHATBOT UX

Experience map – first to next visit

The student journey is a small set of predictable conversational actions, not a maze of invented pages.



Observed vs inferred: the welcome screen and six menu buttons are observed. The ordered journey is a proposed usability map assembled from routes in the JSON; it is not a recording of a single complete student session.

PART I / CHATBOT UX

Learning paths & content taxonomy

Students can study Unit 1 and The Lost World, Chapter 1 through command-driven focus.



VOCABULARY

V1 Core Vocabulary | V2 Academic Vocabulary

GRAMMAR

R1 / R2 Relative Clauses | P1 Participles

CONDITIONALS

C1 Conditionals & Hedging | C2 Inversions

COMPOSITION

S1 Sentence Synthesis | JRE+ writing

COMPREHENSION

L1 Reading & Listening | N1 The Lost World

ALL SKILLS

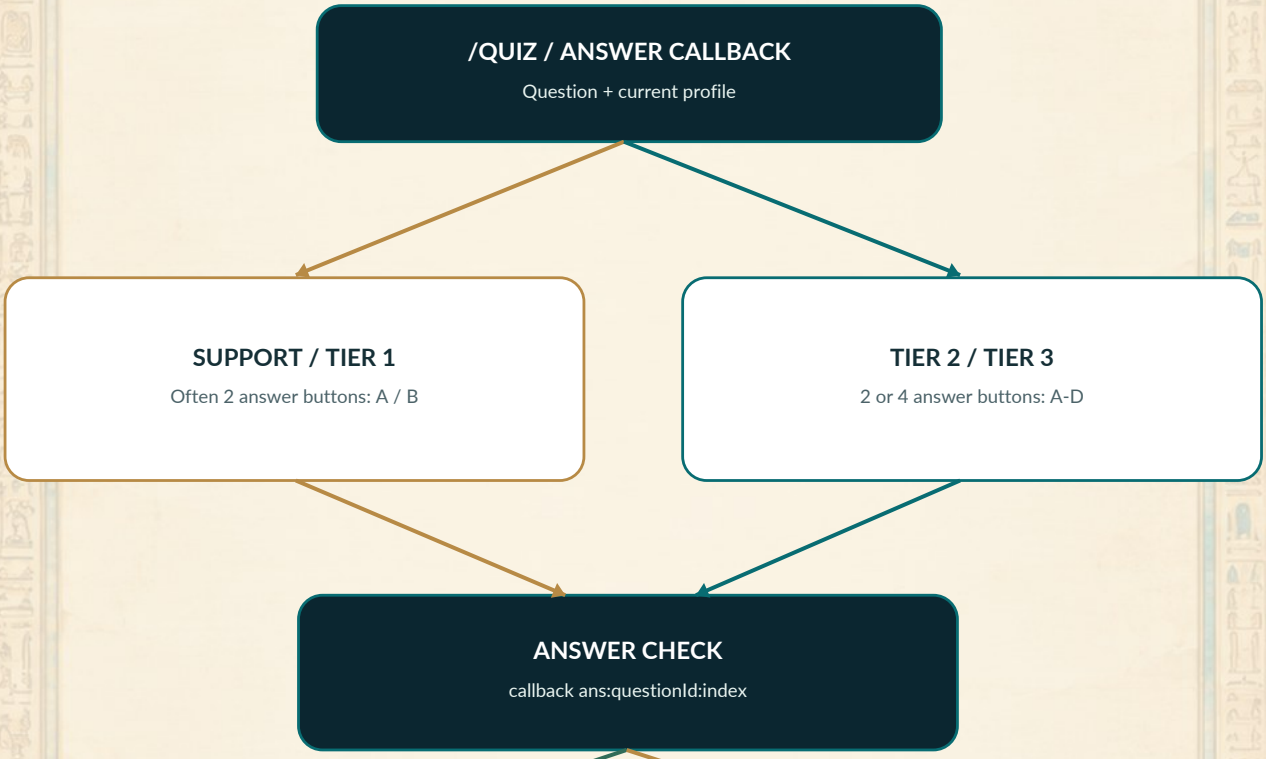
/topic ALL combines the unit practice bank

Scope: Unit 1, Living Well in a Complex World, and Chapter 1 of The Lost World. A bank label does not prove that Qdrant currently contains questions for every bank/tier combination.

PART I / CHATBOT UX

Adaptive quiz: the exact state logic

A code-derived interaction model, illustrated using the real keyboard formats.



SECOND ERROR

Show correct answer; tier down.

SET COMPLETE

After 5 resolved items, evaluate 70% rule.

Production distinction: these are reconstructed states, not production screenshots. Question choices are dynamically shuffled. A stale callback causes a new question rather than grading an obsolete one.

PART I / CHATBOT UX

A gentle correction system

The first wrong answer should feel like a prompt to reason, never a public failure.

A — FIRST-TRY SUCCESS

Correct response + rationale. Consecutive first-try successes can move the learner up.

RECONSTRUCTED RESPONSE

B — FIRST WRONG TRY

Eye of Horus explanation with the answer masked; same choices offered again.

RECONSTRUCTED RESPONSE

C — SECOND WRONG TRY

Correct answer revealed with a rationale; learner is moved down one tier if possible.

RECONSTRUCTED RESPONSE

D — NEXT QUESTION

A Next question button advances to the next practice item; /menu gives a safe exit.

RECONSTRUCTED RESPONSE

Adaptive rule: two consecutive first-try correct answers move up (except at Tier 3). After a second wrong attempt, the tier moves down (except at Support). A 5-question report counts resolved questions, not taps.

PART I / CHATBOT UX

Readiness tiers & student profile

Manual choice and adaptive transitions are distinct controls.

03

CHALLENGE

Higher-order reasoning, complex grammar and synthesis.

02

APPLY

Exam-style application and inference.

01

FOUNDATION

Core concepts, definitions and regular grammar.

00

SUPPORT / D1

Two-choice scaffolding and simpler diagnostics.

Persisted data: student_id, name, email, level, bank, mode, streak, attempt, current question/options, seen IDs, 5-question counters, total counters and per-bank mastery.

/reset: returns to Apply (Tier 2), all topics and chat mode; it does not explicitly erase cumulative total_answered or the seen question history.

PART I / CHATBOT UX

JRE+ paragraph coaching

Socratic writing assistance intends to preserve the student's authorship.

J

JUSTIFICATION

State a debatable claim; avoid absolute wording.

R

REASONING

Explain the claim and consider a counter-view.

E

EVIDENCE

Bring curriculum evidence to support the reasoning.

+

GRAMMAR

Improve expression using relevant Unit 1 structures.

Implementation caveat: /essay selects essay_builder mode and Gemini receives step-by-step instructions. No explicit persistent J/R/E/+ state variable or deterministic step-completion rule is defined in the supplied JSON.

PART I / CHATBOT UX

Conversational tutor & safety

The free-text chat route should guide, redirect and protect.

GEMINI SOCRATIC TUTOR

Scope: Unit 1 and The Lost World (Chapter 1). The system prompt asks one guiding question, keeps responses concise, and avoids direct homework answers.

SHORT-TERM CONTEXT

Per-student memory buffer with a 30-message window is connected to the tutor. A long-term Qdrant retrieval tool is not connected to this agent in the supplied graph.

WINGS OF ISIS

Routes distress to an empathetic response, and cheating/off-topic requests to an educational redirection before agent invocation.

FAILURE RESILIENCE

The Format Socratic Reply code can use a set of deterministic curriculum-specific fallbacks if Gemini output is missing.

REAL UX REQUIREMENTS

Readable mobile messages; predictable /menu exit; clear buttons; safe error copy; no invented claims of certified student-data compliance; appropriate teacher escalation.

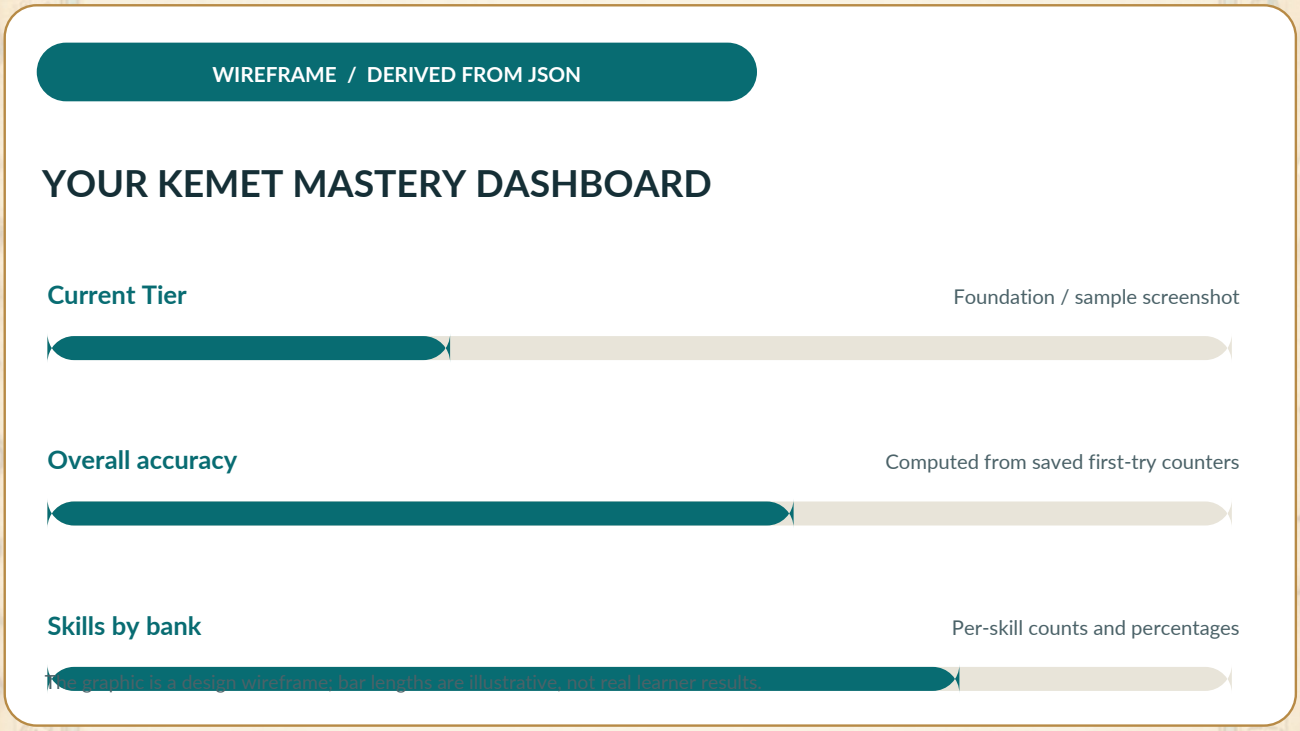
Audit finding: Several source strings contain broken character encoding. Distress and command patterns, especially Arabic variants, need UTF-8 repair and multilingual testing.

PART I / CHATBOT UX

Progress dashboard & reporting

Two channels: immediate progress in Telegram and a formatted report through email.

WIREFRAME / DERIVED FROM JSON



IN TELEGRAM

/progress shows a breakdown of answered items, first-try accuracy, active tier, focus bank, streak and weak-skill recommendation.

IN EMAIL

A report can be prepared on request and at five-question completion. Gmail sends an HTML message and the workflow attempts to attach the PDF.

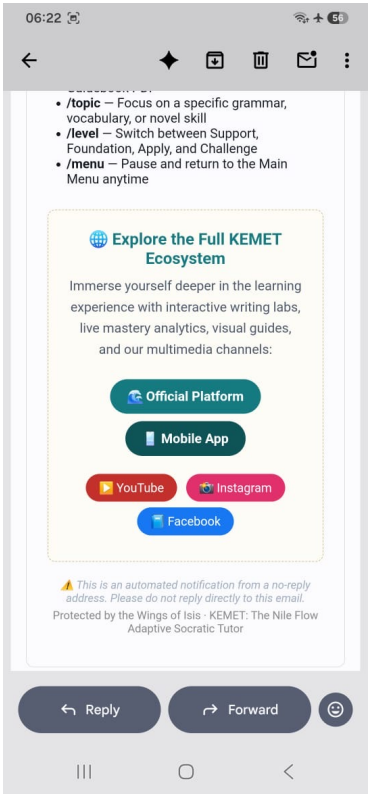
Truthful measurement: the observed account shows 39 questions practised. No measured accuracy or charts were provided; do not present these example bars as actual analytics.

PART I / CHATBOT UX

Guidebook + email touchpoint

The new official banner replaces the old email artwork, not the entire real email interaction.





/GUIDEBOOK

If a stored email exists, prepare the report and PDF. Otherwise ask for an email, then generate a Guidebook email payload.

EMAIL VISUAL HIERARCHY

Official KEMET banner at the top; Guidebook or mastery purpose; student details; report content; ecosystem links; no-reply notice.

Observed historical email: the adjacent screenshot shows the earlier content. It is included only as email-flow evidence; its old header should not be reused.

PART I / CHATBOT UX

Accessibility, ethics & quality review

A beautiful experience must also be legible, predictable and independently testable.

01 CLARITY

Keep authentic Telegram copy and buttons. Avoid generated lookalike screenshots.

02 LENGTH

Break dense /menu copy into scannable sections for mobile usability.

03 FEEDBACK

Distinguish pending, sent, saved and failed actions honestly.

04 LEARNING

Offer hints before answers and a clear way to change tier.

05 SAFETY

Audit Unicode, distress patterns, refusal and human escalation.

06 PRIVACY

Minimize collected personal information and document retention/consent.

07 ACCESS

Test touch target sizing, readability, contrast and command discoverability.

08 ACCURACY

Validate 330-items claim against live Qdrant count; source code shows a smaller seed list.

PART I / CHATBOT UX

Portfolio evidence & decision log

This publication documents the current interface without confusing a visual proposal with implementation.

OBSERVED

The official banner and logo, Telegram main menu screenshot with 39 practised questions, six menu buttons, Gmail historical report screenshot and n8n overview image.

CONFIRMED IN SUPPLIED JSON

Commands /quiz, /essay, /level, /topic, /progress, /guidebook, /reset and /menu; 2/4 option keyboards; profile persistence; quiz feedback; email/memory paths.

CONCEPTUAL / REQUIRES VALIDATION

Reconstructed feedback cards, progress bars, staged JRE+ builder, full 330-question bank, exact PDF attachment and verified email CC, reliable semantic retrieval.

CONVERSATION FIRST. SOURCE-TRUE. LEARNER-CENTRED.

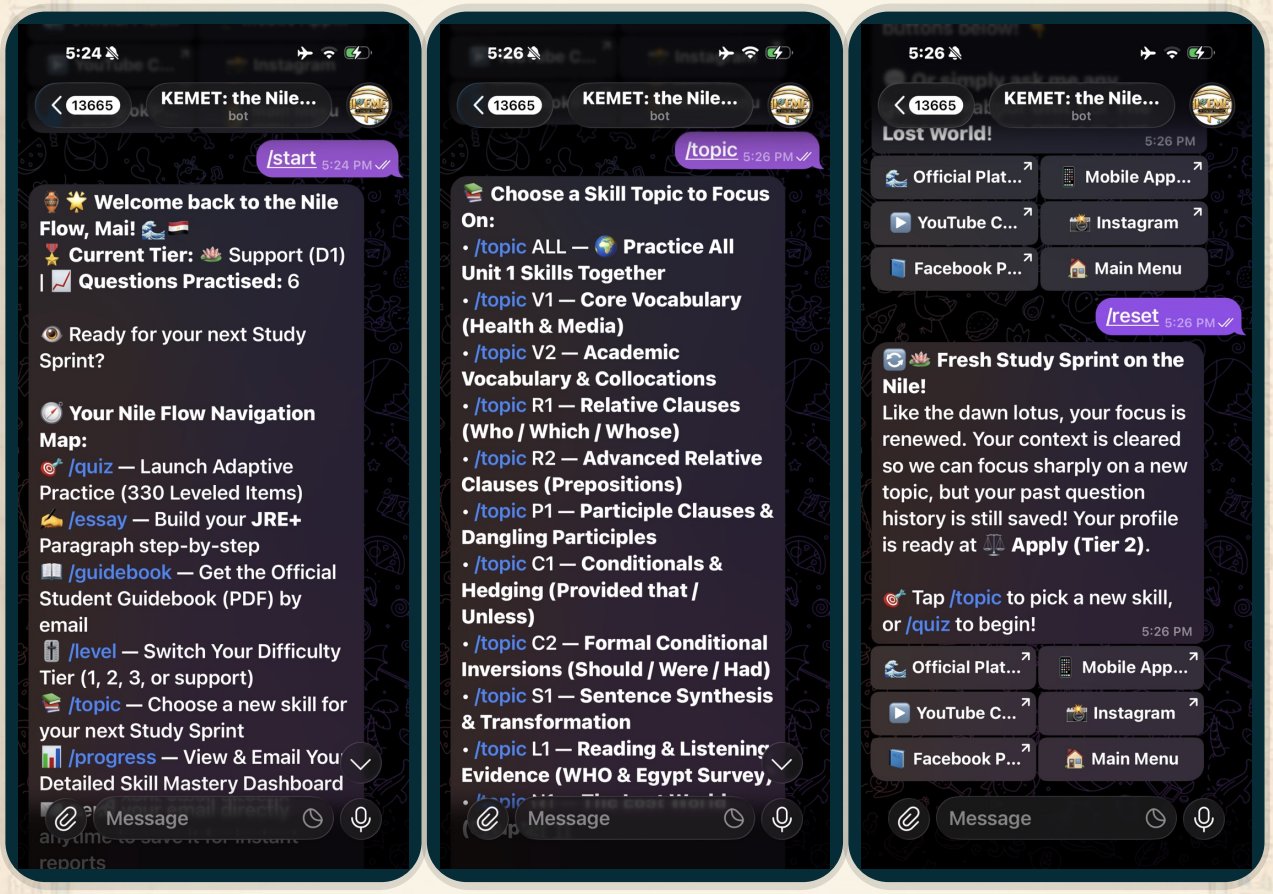
The automation implementation is documented separately.

PART I / CHATBOT UX

Live interaction, not a mockup

Three real interface states, recorded on 30 October 2026. Native Telegram controls are preserved.

01 /start | 02 /topic | 03 /reset



Returning student
Personal greeting, current tier, question count and slash-command navigation.

Choose a topic
ALL, V1, V2, R1, R2, P1, C1, C2, S1, L1 and N1 appear as typed commands.

Start afresh
/reset returns the learner to Apply (Tier 2); previous question history is described as retained.

OBSERVED

These are original Telegram captures. The chatbot is not presented as a separately designed mobile app.

PART I / CHATBOT UX

Question -> hint -> retry

A genuine four-option question shows how the student stays inside one conversation.

01 SELECT A-D | 02 EYE OF HORUS | 03 SET COMPLETED



Adaptive quiz
V1-11 appears at Apply (Tier 2), with A, B, C, D and Main Menu buttons.

First-error scaffold
The bot masks the answer, offers a hint and asks the student to try again.

After five questions
A real 20% set result is shown; the screen reports teacher notification.

UX LOGIC

Answer choice is tapped as an inline button, not typed into a separate quiz screen.

PART I / CHATBOT UX

Support tier in action

Two-choice scaffolding and gradual release, as shown in the real bot.

SUPPORT / A-B | HINT AND RETRY | EXPLANATION



Accessible branching
D1-01 offers A) who and B) which, reducing choice complexity.

Guided attempt
An Eye of Horus hint invites the learner to revisit the same item.

Gradual release
The bot explains the correct form and displays the five-question set outcome.

DESIGN PRINCIPLE

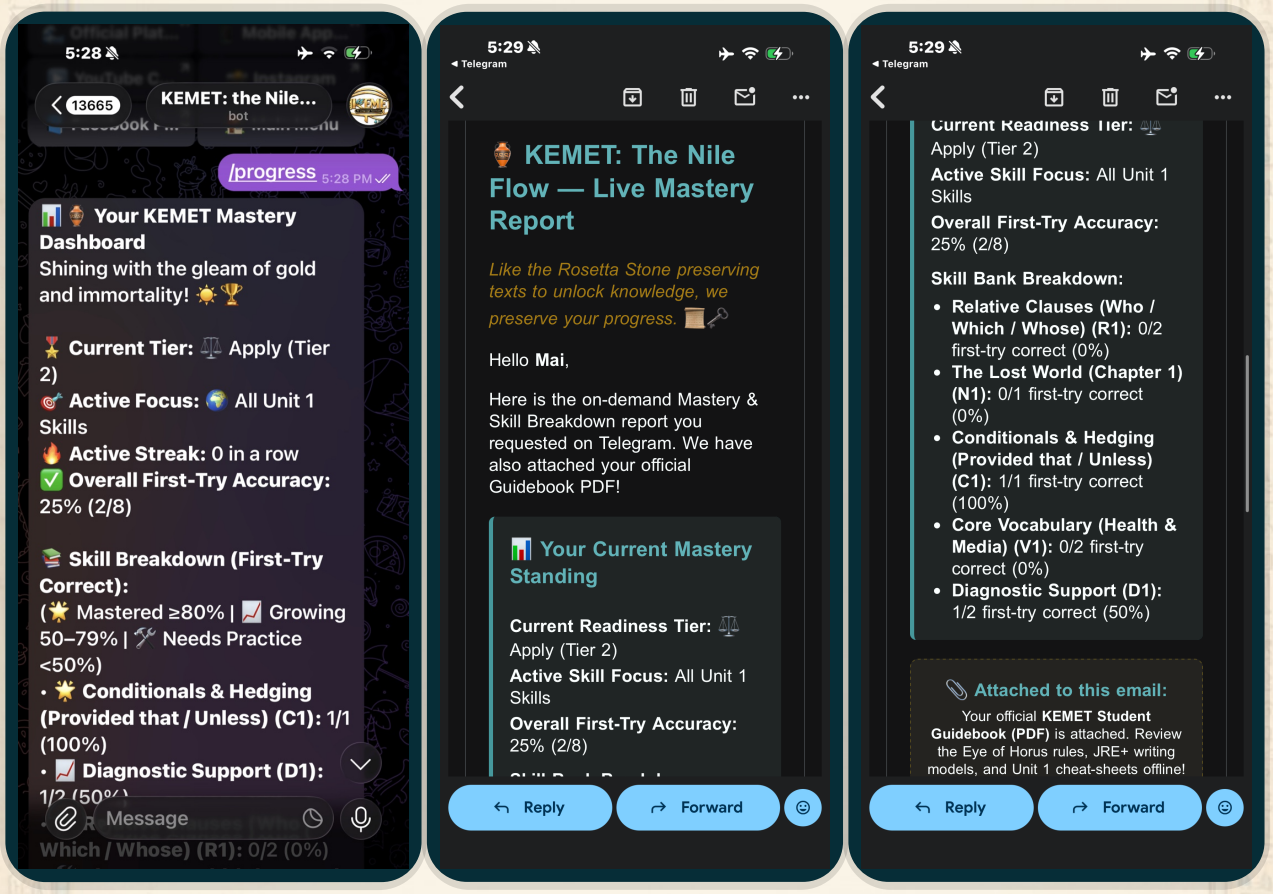
Avoid redesigning the Telegram question/keyboard: clarity comes from progressive support, not invented controls.

PART I / CHATBOT UX

Mastery across two channels

The Telegram response is paired with a delivered Gmail view captured during the same demonstration.

TELEGRAM /progress | GMAIL / REPORT | SKILL BREAKDOWN



In-chat dashboard
Example: Apply Tier 2, all Unit 1 skills, 25% first-try accuracy (2/8).

Email delivered
The Gmail message is visible in the user-supplied capture, with a mastery summary.

Detailed evidence
Skill-level results are presented alongside a Guidebook PDF attachment notice.

EVIDENCE LIMIT

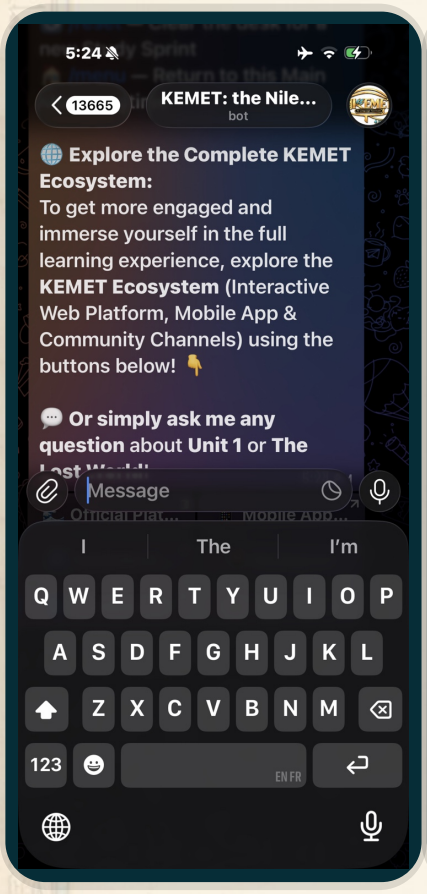
One delivered report demonstrates a successful example. It does not establish delivery reliability for every learner.

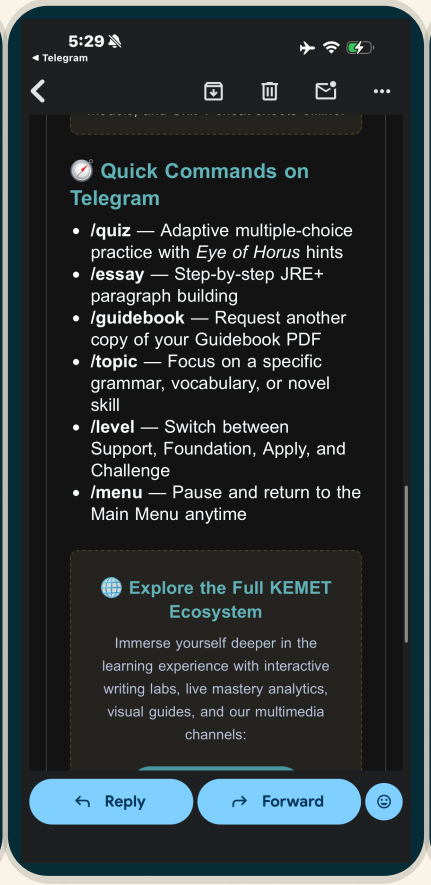
PART I / CHATBOT UX

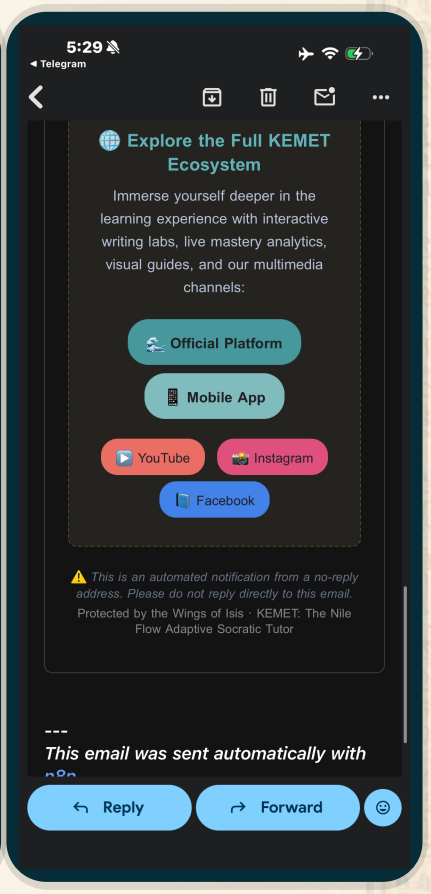
Discovery, commands & ecosystem

Small navigation details make the conversational experience usable beyond quiz questions.

COMPOSE MESSAGE | EMAIL COMMANDS | EMAIL ECOSYSTEM







Native input
Telegram retains its standard composer and keyboard; students can type questions.

Quick reference
The actual email includes /quiz, /essay, /guidebook, /topic, /level and /menu.

Integrated links
Official Platform, Mobile App, YouTube, Instagram and Facebook appear in the email.

VIDEO REVIEW

The supplied 3m56s recording also shows navigation from Telegram to other KEMET channels; external pages are separate services.